

Workshop: Smart City Analysis and K-means Segmentation

Data Science and Causality for Smart Cities

Workshop purpose

Analyze the cities in the *Smart City Index* dataset as complex urban systems, not only as technological rankings. The workshop follows the logic of the infographic *The DNA of a Smart City: Beyond Technology* and the course presentation *Finding the Signal in the Noise*: identify the urban problem, filter noise, explore the signal, discover hidden patterns with K-means, and communicate actionable insights.

Final deliverable

The only deliverable is a **maximum 7-minute oral presentation**. No written report is required. Each team must use visual evidence and explain the results clearly for an urban decision-making audience.

1 Dataset and conceptual frame

Dataset: https://raw.githubusercontent.com/Andres1984/Data-Analysis-with-R/refs/heads/master/Bases/Smart_City_index.csv

A smart city is not simply a city with sensors, dashboards, or digital platforms. A city becomes smart when technology supports human, social, environmental, institutional, and economic needs. For this reason, the analysis must focus on the six pillars of urban smartness:

Dimension	Meaning for urban analysis
Smart Mobility	Sustainable, safe, accessible, and integrated transportation systems.
Smart Environment	Resource management, pollution reduction, sustainability, and natural capital.
Smart Government	Transparency, digital public services, participation, and policy coordination.
Smart Economy	Innovation, productivity, entrepreneurship, and competitiveness.
Smart People	Education, inclusion, human capital, civic capacity, and participation.
Smart Living	Quality of life, health, safety, housing, culture, and everyday services.

Important restriction. K-means must be estimated using **only the six smart city dimensions**. The variables `SmartCity_Index` and `SmartCity_Index_relative_Edmonton` must be **excluded from clustering**. They are analyzed only after clustering to interpret and validate the city profiles.

2 Guiding research question

Each team must answer the following question:

Do cities become “smart” through the same urban development pattern, or do different groups of cities show different combinations of strengths and weaknesses across mobility, environment, government, economy, people, and living conditions?

Suggested hypothesis: cities with similar global smart city scores may have very different internal profiles. Therefore, a ranking alone may hide different urban models.

3 Workshop tasks

Task 1. Understand the data and identify noise

Inspect the dataset before modeling. Identify the number of cities, countries, variables, missing values, duplicated observations, numeric variables, and categorical variables. Explain what kind of “noise” may affect the analysis: scale differences, outliers, missing values, redundancy, or conceptual overlap.

Questions to answer orally:

1. What is the unit of analysis: city, country, or indicator?
2. Are the six dimensions comparable, or do they require standardization?
3. Why should the two aggregate index variables not be used as inputs for K-means?

Task 2. Analyze the six smart city dimensions

Explore the distribution, variation, and relationships among the six smart city pillars. The objective is to discover the shape of the urban signal before applying clustering.

Minimum visual evidence:

1. A comparative plot of the six dimensions.
2. A correlation heatmap among the six dimensions.
3. A profile plot for selected cities that shows strengths and weaknesses across pillars.

Guide Questions

1. Which dimension shows the highest and lowest average performance?
2. Which dimension varies the most across cities?
3. Are the leading cities balanced, or do they depend on a few strong pillars?

Task 3. Analyze the aggregate variables

Analyze `SmartCity_Index` and `SmartCity_Index_relative_Edmonton` separately. These variables summarize global performance and relative position, but they should not replace the six-dimensional analysis.

Minimum visual evidence:

1. Top 10 and bottom 10 cities by `SmartCity_Index`.
2. Distribution of `SmartCity_Index_relative_Edmonton`.
3. Identification of cities above and below Edmonton's reference level.

Guide Questions

1. Which cities lead the global index and which are at the bottom?
2. What does the Edmonton-relative variable add to the interpretation?
3. Does a high global index imply balance across all six pillars?

Task 4. Apply K-means only to the six dimensions

Estimate K-means using the standardized six smart city dimensions. Do not include `SmartCity_Index`, `SmartCity_Index_relative`, city names, or country names as clustering inputs.

Minimum analytical decisions:

1. Justify the selected number of clusters using an interpretable criterion such as elbow plot, silhouette, or conceptual usefulness.
2. Assign each city to a cluster and profile each group.

Guide Questions:

1. How many clusters were selected and why?
2. What are the strengths and weaknesses of each cluster?
3. Which cities best represent each cluster?
4. How do clusters differ in `SmartCity_Index` after the clustering is completed?

4 Expected presentation output

Each presentation should include the following elements:

1. One clear research question.
2. One explanation of the dataset and the six smart city dimensions.
3. One short diagnosis of data noise and preprocessing decisions.
4. One descriptive analysis of the six dimensions.
5. One analysis of `SmartCity_Index` and `SmartCity_Index_relative_Edmonton`.
6. One K-means segmentation based only on the six dimensions.
7. One interpretation of each cluster using meaningful urban labels, for example: *balanced leaders*, *mobility-oriented cities*, *governance-driven cities*, *environmental laggards*, or *emerging smart cities*.
8. One policy recommendation per cluster.
9. One final message explaining what was learned beyond the global ranking.